Databricks SQL Project Report

Dataset: Airbnb Dataset (Airbnb Listings in New York City)

1. Project Overview:

This project analyzes Airbnb dataset using Databricks SQL including DDL, DML and visualization to understand pricing and neighbourhood trends.

2. Dataset Used:

Airbnb Dataset – listing details like name, neighbourhood, price, property type, review scores.

3. Schema Used:

Catalog: workspace | Schema: default | Table: airbnb

4. Steps Performed:

❖ Data Preview

Condition: SELECT * FROM airbnb LIMIT 100

Description: Displays a sample of the dataset to understand structure, columns, and initial data quality.

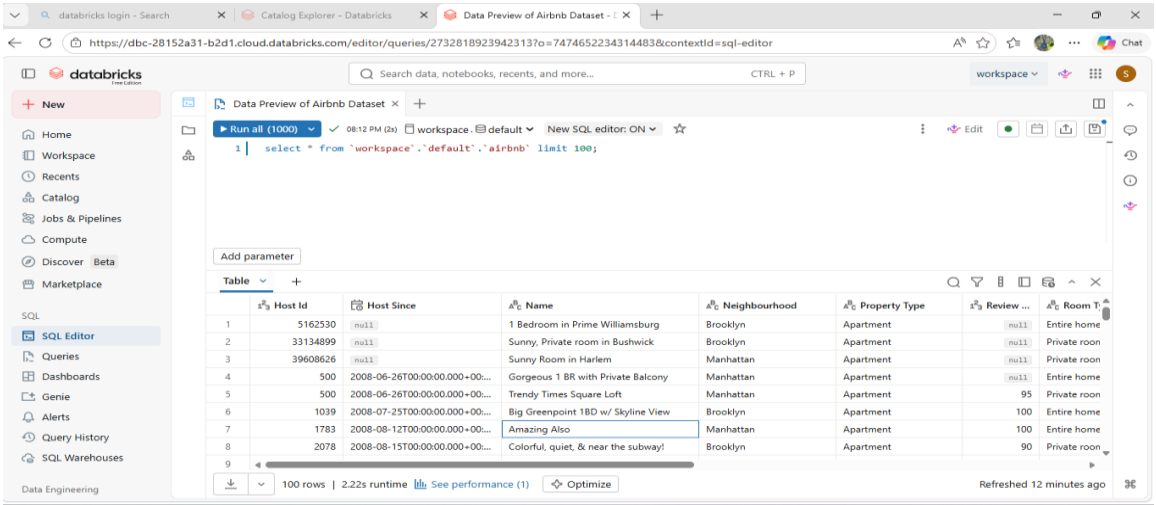

Fig. 1: Dataset preview (airbnb table sample)

DDL (Data Definition Language)

❖ DDL 1 - High Rated Listings

Condition: WHERE review_scores_rating > 85

Description: Filters and identifies high-rated listings based on review scores to analyze top-performing properties.

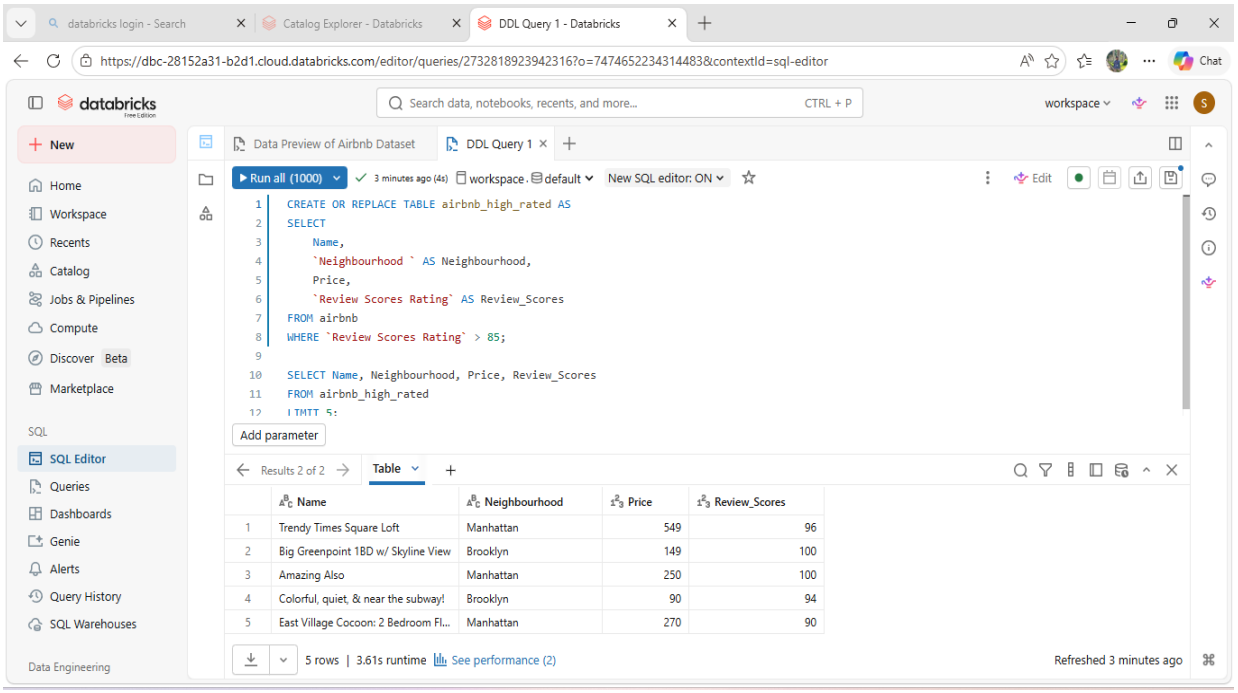

Fig. 2: High-rated listings

❖ DDL 2 - Mid Price Range

Condition: WHERE price BETWEEN 100 AND 300

Description: Extracts listings within the mid-price range to study moderately priced properties.

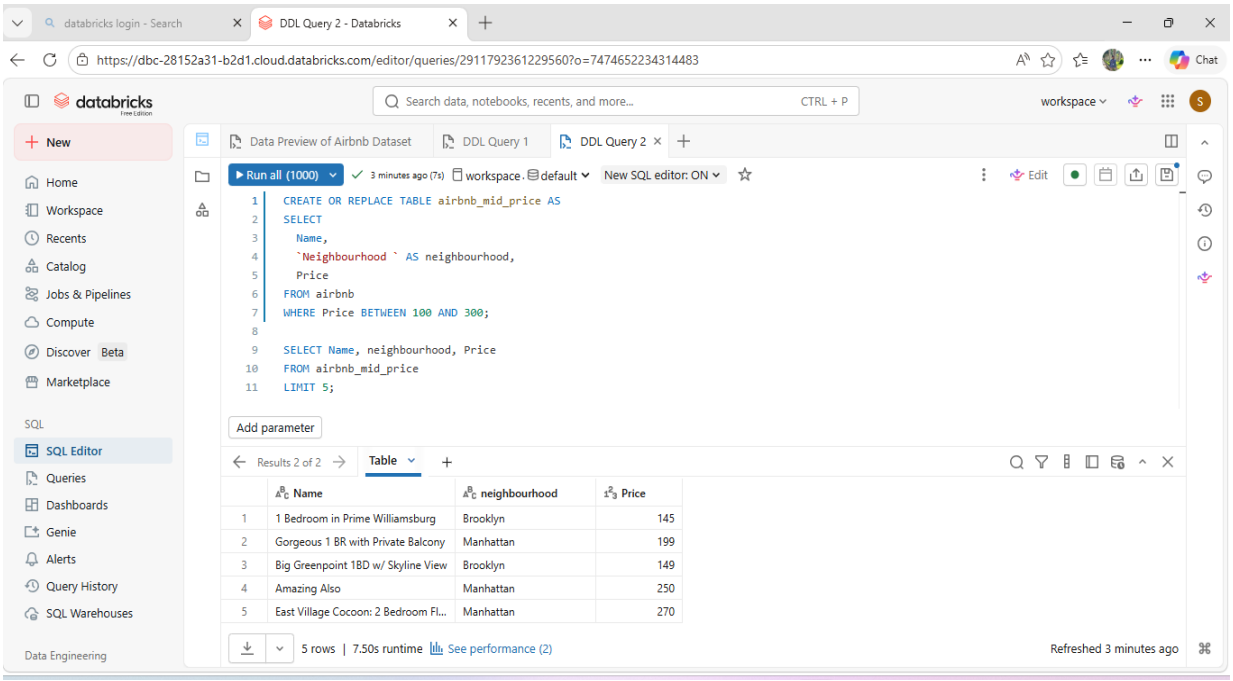

Fig. 3: Mid-range listings by price

❖ DDL 3 - Price Categorization

Condition: CASE statement for price categorization (low, medium, high)

Description: Categorizes listings into price groups (low, medium, high) using conditional logic for better analysis.

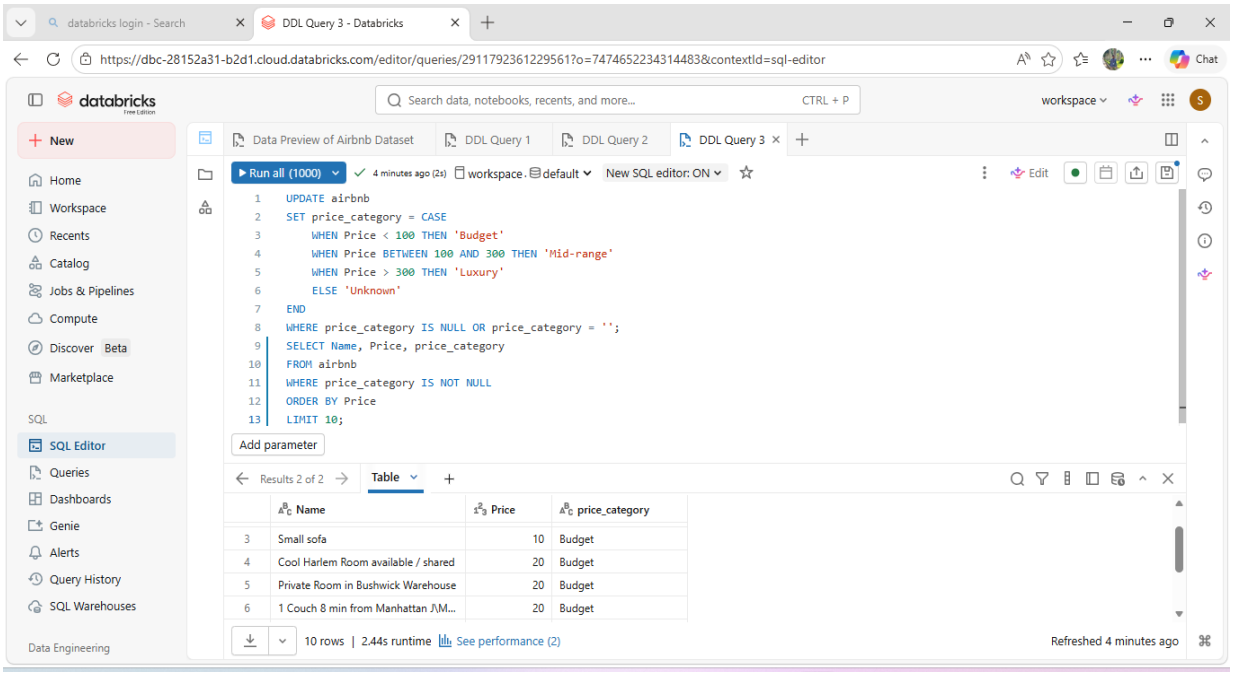

Fig. 4: Price category classification using case

❖ DDL 4 - Group By

Condition: GROUP BY neighbourhood with COUNT(*) aggregation

Description: Groups listings by neighbourhood to summarize key metrics like average price and listing count.

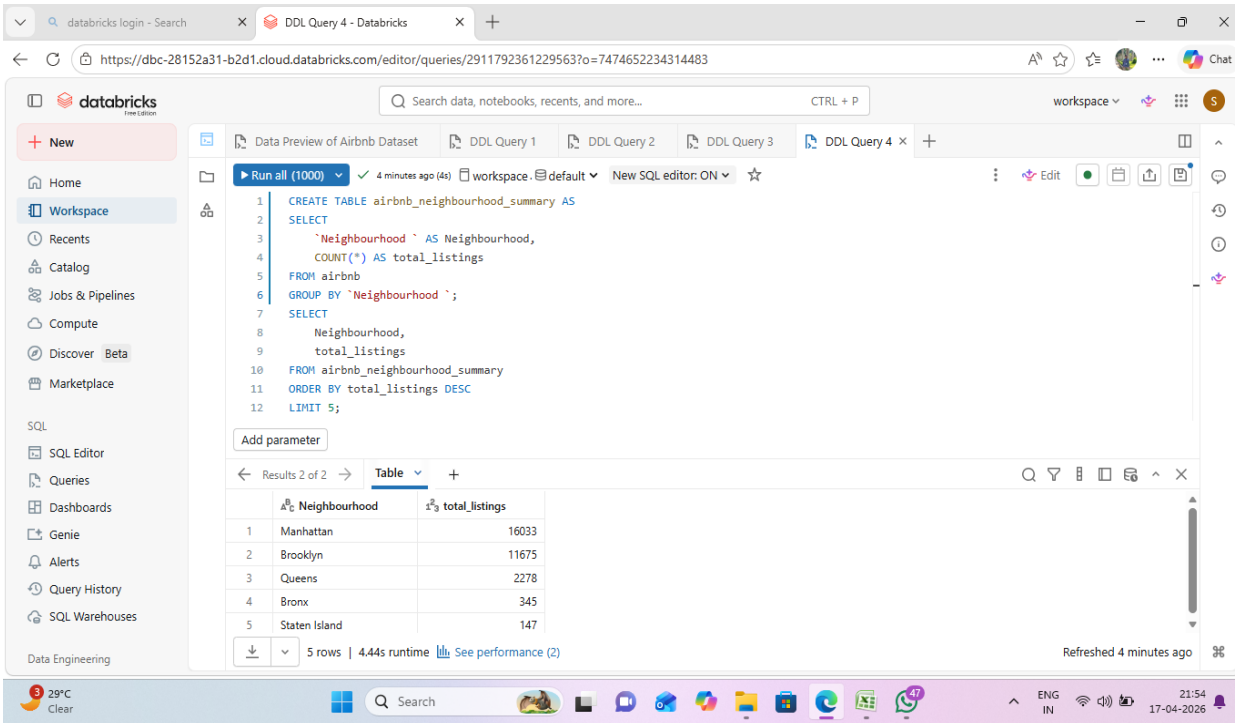

Fig. 5: Neighbourhood-wise summary

❖ DDL 5 - Sorting

Condition: ORDER BY price DESC

Description: Sorts listings in descending order to highlight highest values such as price or ratings.

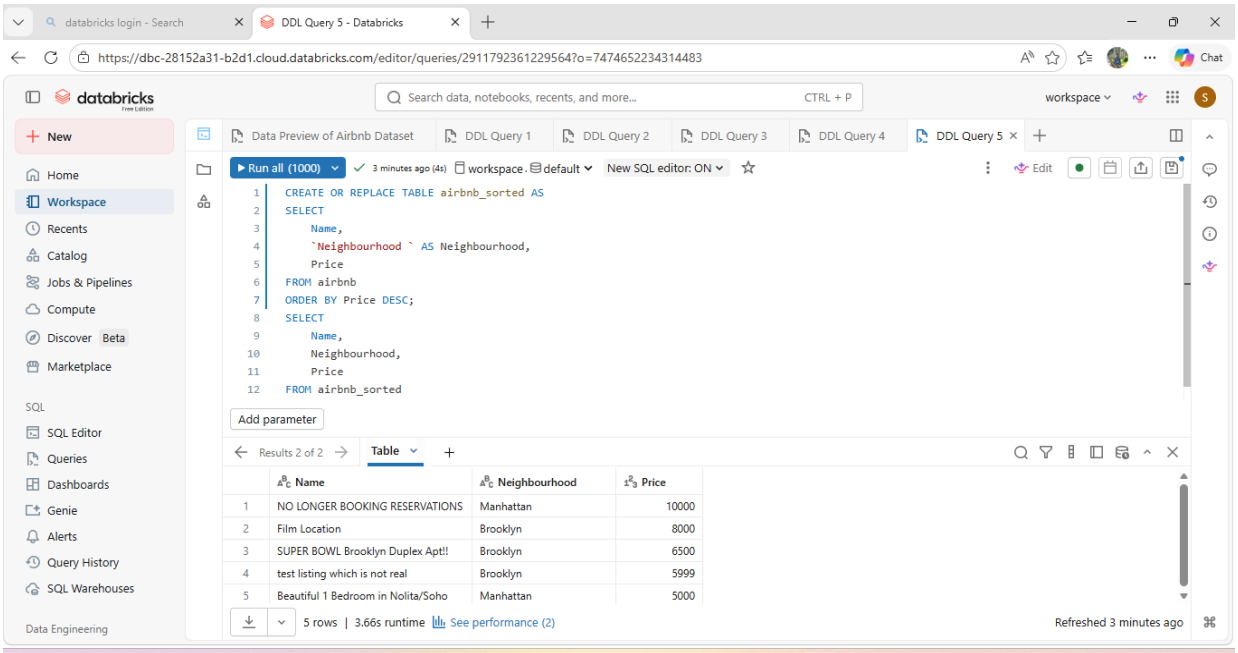

Fig. 6: Sorted listings in descending order

❖ DDL Visualization

Condition: AVG (price) GROUP BY neighbourhood

Description: Visualizes average price across categories or neighbourhoods to compare pricing trends.

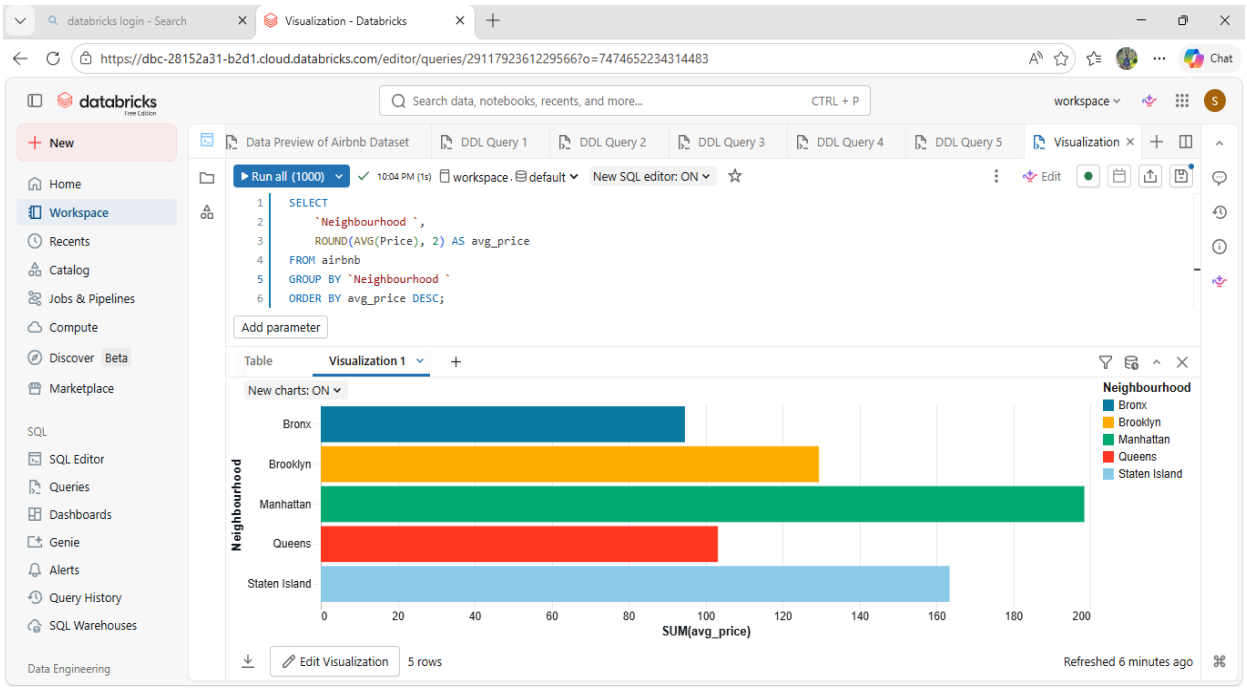

Fig. 7: Average price comparison

DML (Data Manipulation Language)

❖ DML 1 - Insert

Condition: INSERT INTO airbnb VALUES

Description: Adds new records into the dataset to simulate data entry or updates.

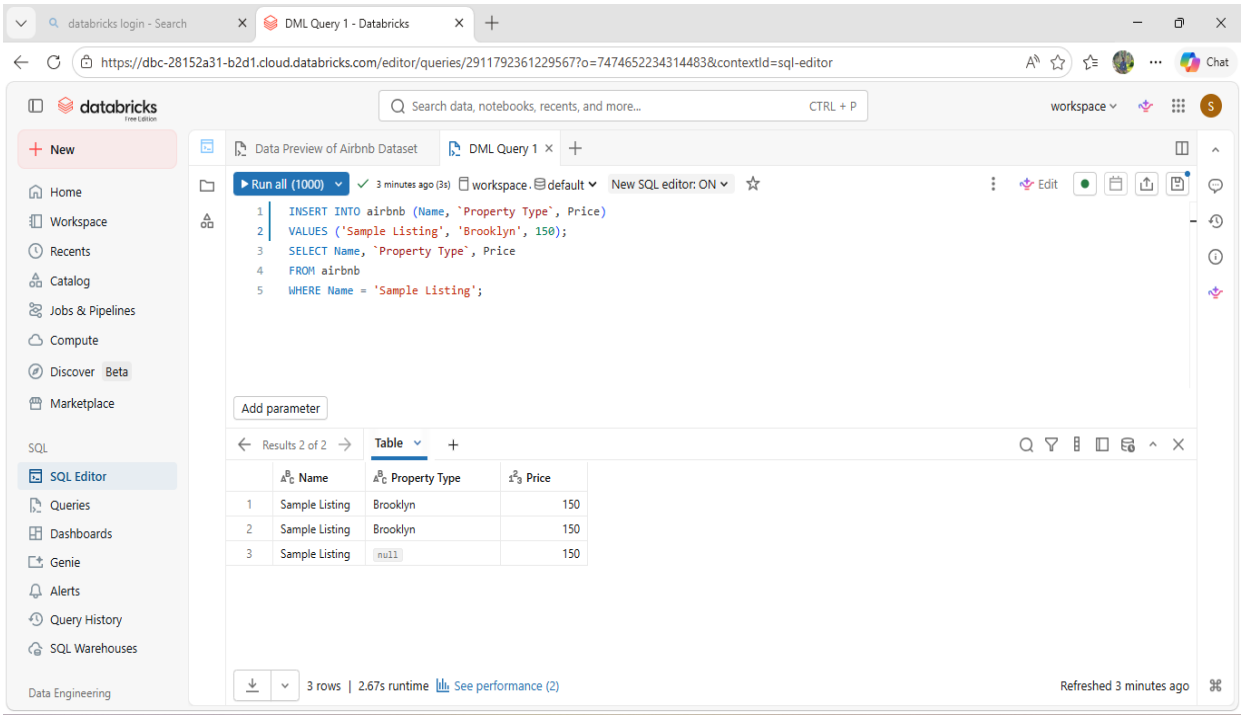

Fig. 8: Data insertion operation

❖ DML 2 - Update

Condition: UPDATE airbnb SET column name = value WHERE condition

Description: Modifies existing records to correct or update listing information.

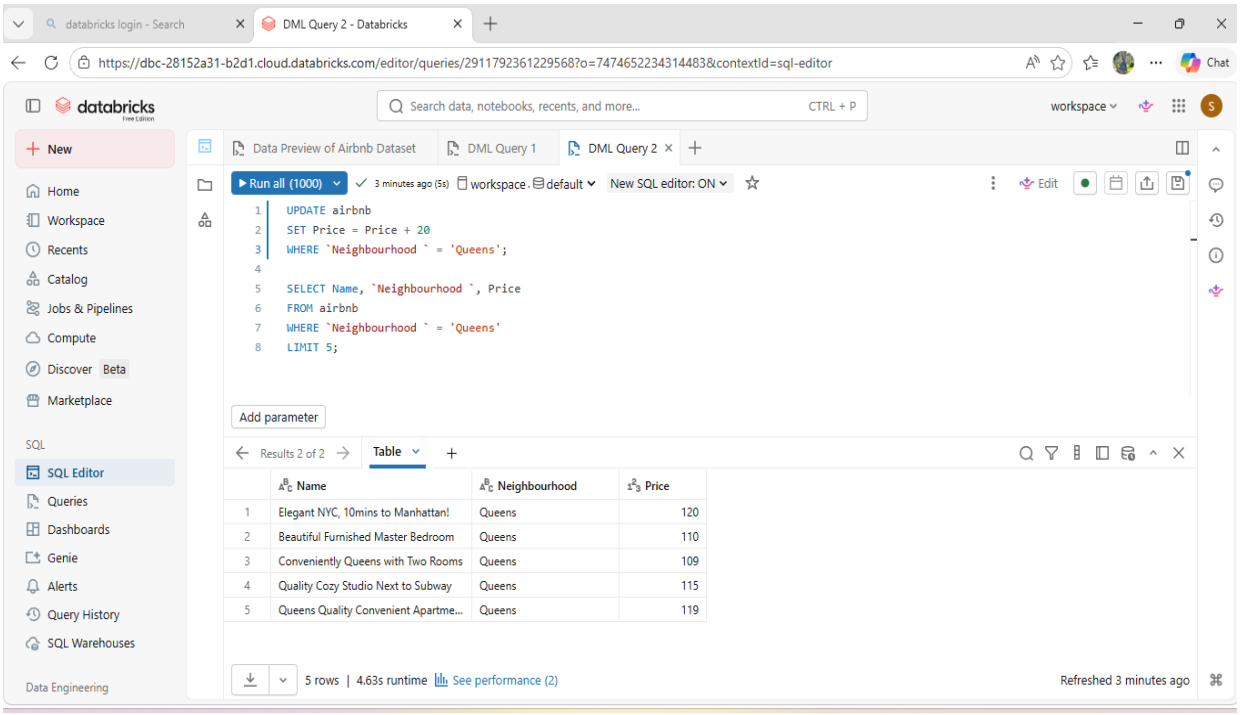

Fig. 9: Data update operation

❖ DML 3 - Aggregation

Condition: GROUP BY neighbourhood with aggregation functions

Description: Performs aggregation on data to analyze grouped insights such as totals or averages.

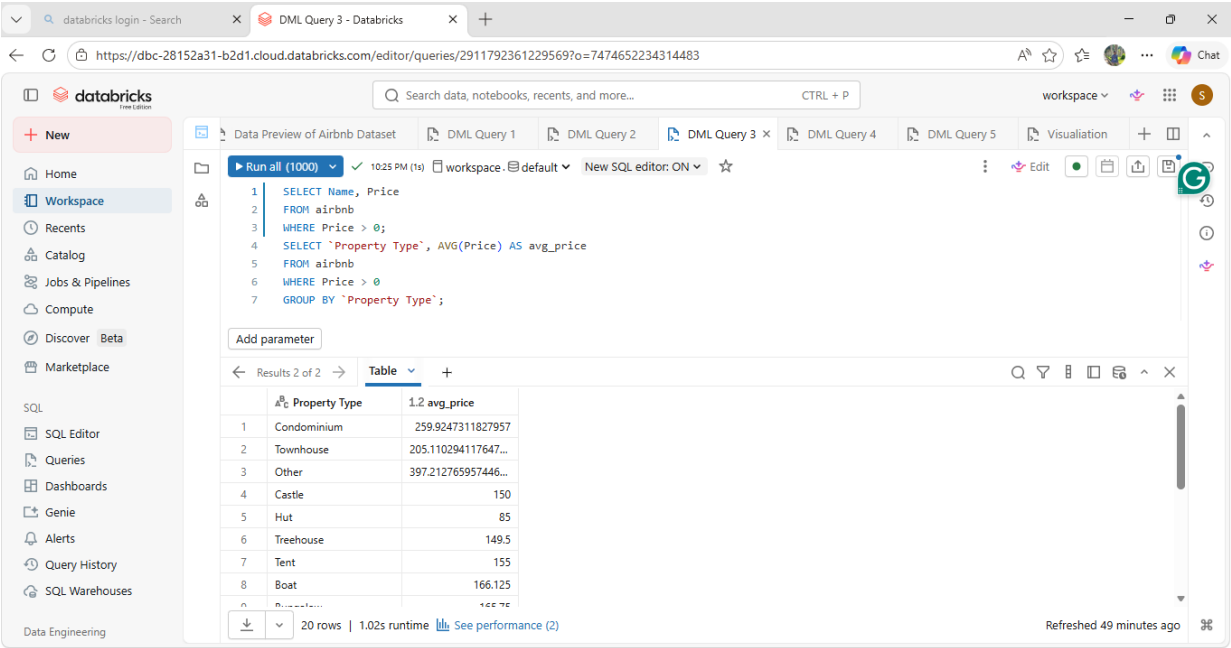

Fig. 10: Aggregated results

❖ DML 4 - High Price Filter

Condition: Price > 300

Description: Filters high-priced listings to analyze premium property trends.

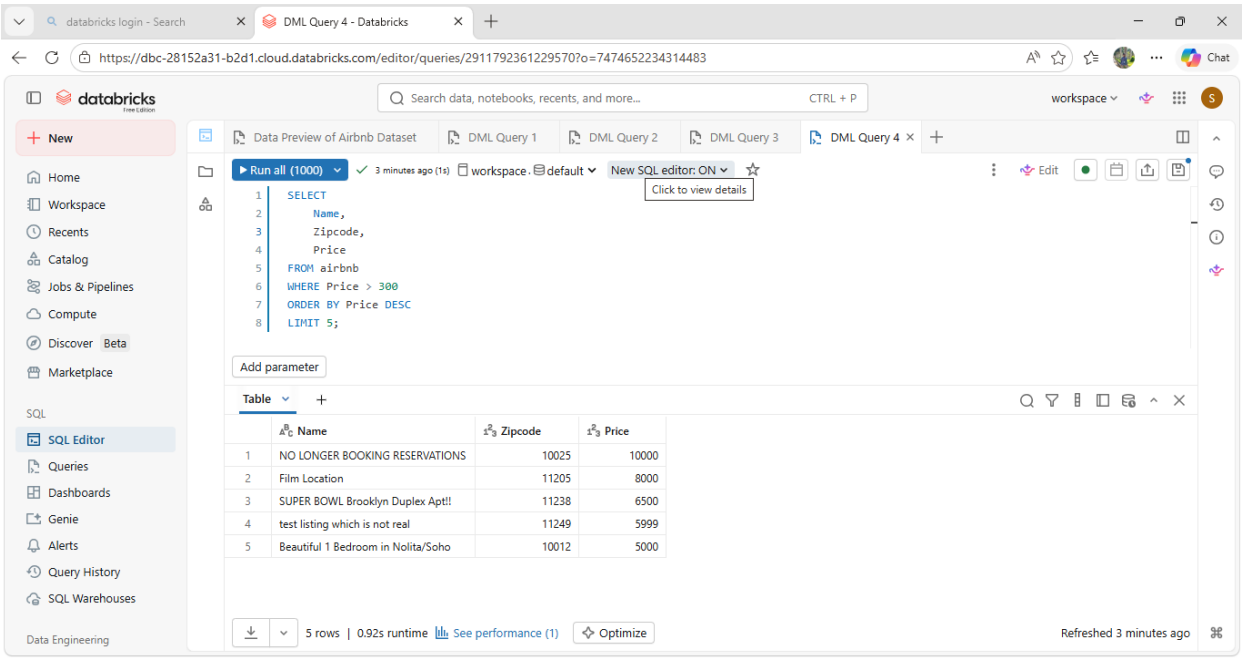

Fig. 11: High-price listings filter

❖ DML 5 - Count

Condition: SELECT COUNT (*) FROM airbnb

Description: Counts the total number of listings to understand dataset size or grouped distribution.

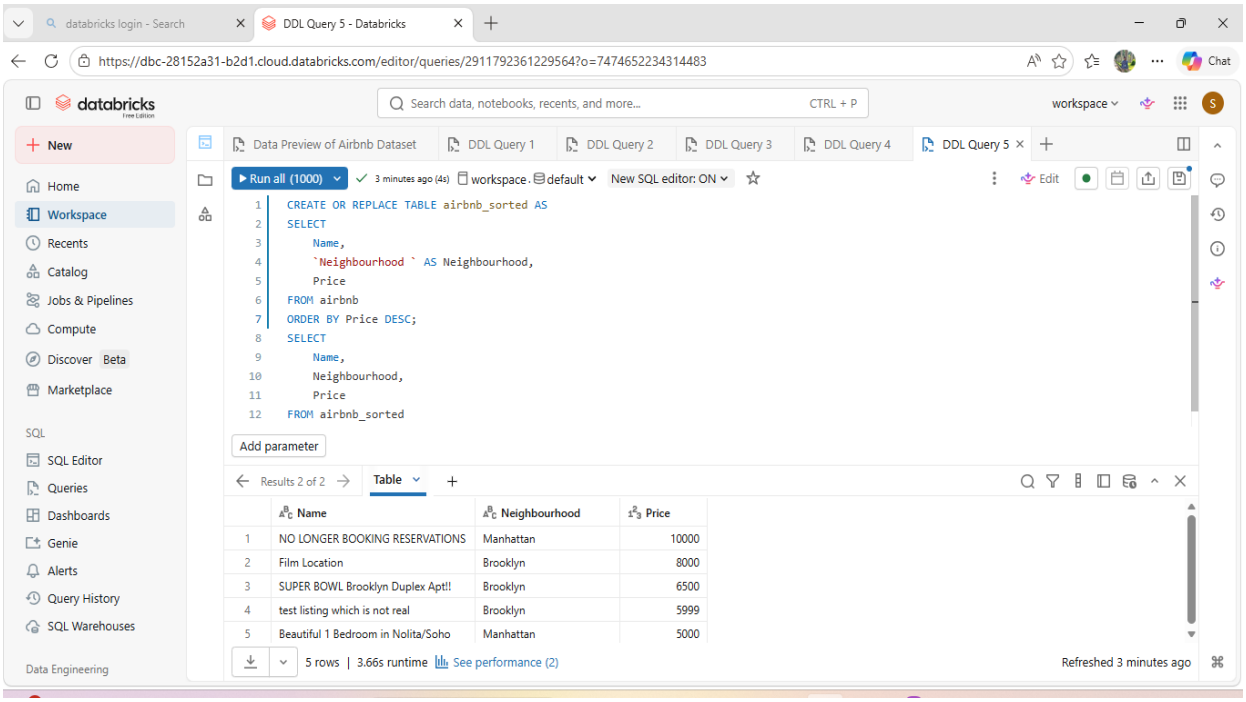

Fig. 12: Total listing count

❖ DML Visualization

Condition: AVG (price) GROUP BY neighbourhood

Description: Represents average price visually to identify patterns and comparisons easily.

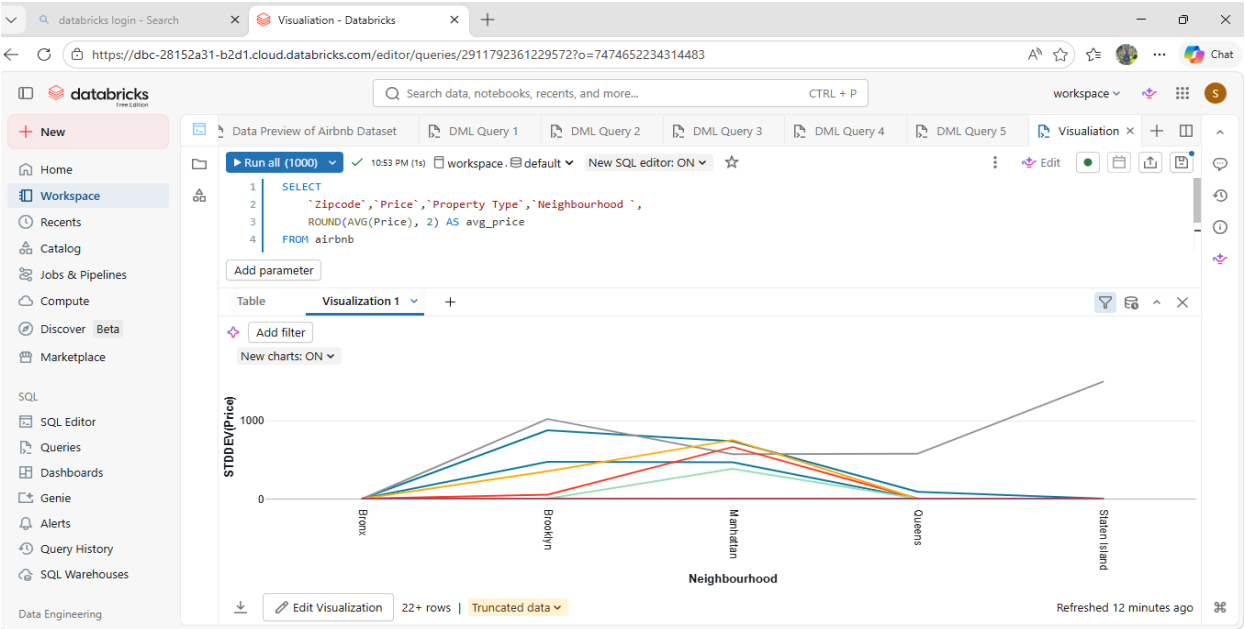

Fig. 13: Average price visualization

5. Conclusion:

This project demonstrated the effective use of Databricks for handling data using DDL and DML operations. Tables were created to filter high-rated listings, mid-price properties, and sorted datasets. Data was also updated to categorize listings based on price range, and aggregation was performed to analyze neighbourhood-wise trends. These operations helped in understanding data structure and extracting meaningful insights from the dataset.

6. Result:

The project was successfully completed with proper execution of SQL queries. New tables such as high-rated listings, mid-price listings, and neighbourhood summary were created. Data updates like price categorization were applied, and queries were used to retrieve and analyze results. The final outputs and visualizations clearly showed price distribution and neighbourhood insights.

7. Key Outcomes:

Created tables for:

- High-rated listings (Review Score > 85)
- Mid-price listings (Price between 100 and 300)
- Sorted listings based on price
- Neighbourhood-wise listing summary

Performed data updates:

- Categorized listings into Budget, Mid-range, and Luxury using CASE statement

Executed data analysis queries:

- Retrieved filtered records
- Calculated average price by neighbourhood
- Counted listings per category

Improved understanding of:

- SQL query writing and execution
- Data filtering, grouping, and sorting
- Real-world dataset handling in Databricks
- Data aggregation and visualization concept

"This project reflects practical understanding of SQL operations and real-world data analysis using Databricks."

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